

Deliverable D13 · Work Package WP4

Verification of the proposed concept

Report on the experimental proof of concept

Project identification:

Project title: A Statistical Approach to Monitor Quantisation in Neural Network Training

Project acronym: SAMQ-NN

Project code: 09I03-03-V04-00562

Programme: Recovery and Resilience Plan of the Slovak Republic

Component: 9. More effective governance and strengthening of the funding of research, development and innovation under the Recovery and Resilience Plan of the Slovak Republic

Investment: 3. Excellent science

Aid scheme: State aid scheme to support research, development and innovation under Component 9 of the Recovery and Resilience Plan of the Slovak Republic, No. SA.106633

Project start: 06/2024

End of implementation: 05/2026

Principal investigator: Ing. Roman Budjač, PhD.

Summary

Deliverable D13 experimentally proves the concept of monitoring quantisation by means of a statistical similarity index of the weight distribution, proposed in deliverable D12. On four small datasets (digits, Fashion-MNIST, MNIST, CIFAR-10) and on large pretrained models (MobileNetV2 and ResNet50 evaluated on ImageNet-V2), it was confirmed that the index monotonically tracks the severity of quantisation ($\rho \approx -0.98$), predicts the drop in accuracy, and is stable during QAT as well as reproducible across datasets and architectures. All four validation criteria set out in D10 are met, and the quantisation itself brings a real saving in model size (8-bit is approximately 4× at practically lossless accuracy). At the same time the key limitation is named honestly: a global weight index does not capture degradation that does not show up in the weight distribution (MobileNetV2 at 4-bit). This fulfils Objective 6, i.e. the concept is proven on standard datasets and evaluated against the FP32 reference as well as against the criteria from D10.

1. Aim and context

Deliverable D13 experimentally proves the concept implemented in deliverable **D12**: monitoring the quantisation of a neural network by means of a statistical similarity index of the weight distribution (Jensen-Shannon divergence between the weight distribution of the quantised and the full-precision model). The notebook directly fulfils **Objective 6** (“Prove the concept on standard datasets and evaluate results and potential benefits and limitations of the proposed solution”). The verification is carried out on standard datasets, against four validation criteria set out in **D10**, and against the full-precision (FP32) reference.

2. Method and scope of the verification

- **Index:** JS divergence between the weight distribution of the quantised and the FP32 model; the quantisation is per-channel (a separate scale per output channel).
- **Small models:** digits, Fashion-MNIST, MNIST, and CIFAR-10, each across 5 independent initialisations (seeds).
- **Large models:** MobileNetV2 and ResNet50 pretrained on ImageNet, evaluated on a validation subset of ImageNet-V2 (~5,000 images).
- **Scope:** bit widths 2, 4, 8, 16, and 32; post-training quantisation (PTQ) as well as the dynamics during genuine QAT.

3. Results and validation against the criteria from D10

All four criteria from D10 are met on the obtained data:

Validation criterion (D10)	Status	Key measured value
1. Index threshold value ($\tau = 0.01$)	met	The index stays low precisely at ≥ 8 bits (accuracy preserved); JS at 8 b ≈ 0.000
2. Correlation of the index with accuracy	met	$\rho(\text{bits, JS}) = -0.98$ across datasets; a higher index corresponds to lower accuracy and higher loss
3. Stability across epochs (QAT)	met	Standard deviation of the index over epochs ≤ 0.003 ; the weights remain genuinely quantised
4. Reproducibility	met	The same behaviour on 4 small datasets as well as on MobileNetV2 and ResNet50 (ImageNet-V2)

Efficiency and compression (quantified in three ways):

- Theoretically: 16-bit = 2 $\times$, 8-bit = 4 $\times$, 4-bit = 8 $\times$ smaller model; 8-bit at practically lossless accuracy.
- Real int8/TFLite export of the small model: 57.9 kB $\rightarrow$ 21.5 kB (2.70 $\times$); for larger models the ratio approaches the theoretical 4 $\times$.
- Latency (CPU, 1 sample): FP32 = 0.004 ms vs. INT8 = 0.007 ms. On an ordinary CPU INT8 does not become faster (the overhead is offset by an optimised FP32); a significant speedup comes only with hardware that has INT8 acceleration (mobile and edge NPUs).
- ImageNet: ResNet50 at 8-bit is practically lossless (top-1 0.562 vs. FP32 0.563); MobileNetV2 at 4-bit collapses (top-1 0.002).

4. Benefits and limitations

Benefits. The index is cheap (its overhead is negligible relative to a single forward pass of the network) and is computed continuously during training; it is dataset-agnostic and independent of the specific metric (KL/JS/cosine are strongly correlated); it works both as a predictor (selecting the bit width before training) and as a monitor (stable during QAT). The per-layer breakdown identifies the most sensitive layers (the first convolution and the output layer), which opens the way to mixed precision.

Limitation (named honestly). A global weight index does not capture degradation that does not show up in the weight distribution: MobileNetV2 collapses at 4-bit, yet the global JS remains negligible ($\approx 1e-4$). This defines the direction of further work, namely complementary monitoring at the level of layers and activations. In addition, the index threshold needs to be calibrated empirically for a given architecture, so it is recommended to track it together with accuracy.

5. Conclusion

The experiments confirm the proposed concept: the statistical similarity index monotonically tracks the severity of quantisation ($\rho \approx -0.98$ across the small datasets, -1.00 on ImageNet), predicts the drop in accuracy, is stable during QAT, and is reproducible across datasets and architectures, while the quantisation itself brings a real saving in model size. This fulfils Objective 6: the concept is proven on standard datasets and evaluated against the FP32 baseline as well as against the criteria from D10. The results form a direct basis for the written summary (D15) and the final report (D17). The full measurements, plots, and source code are in the accompanying D13 notebook.